PROBLEM 26

Heavy Particle on a Rotating Half-Circumference

Original statement · Problems.pdf, p. 14

Semicircular Rotating Wire

A heavy point particle P of mass m is constrained to move without friction along the half-circumference wire of radius a shown in the figure. The wire containing the point particle rotates around the OY axis with constant angular velocity $\boldsymbol{\omega} = \Omega\mathbf{j}$. Initially, the particle is located at $x = 0$ with velocity v_0 along the positive x axis.

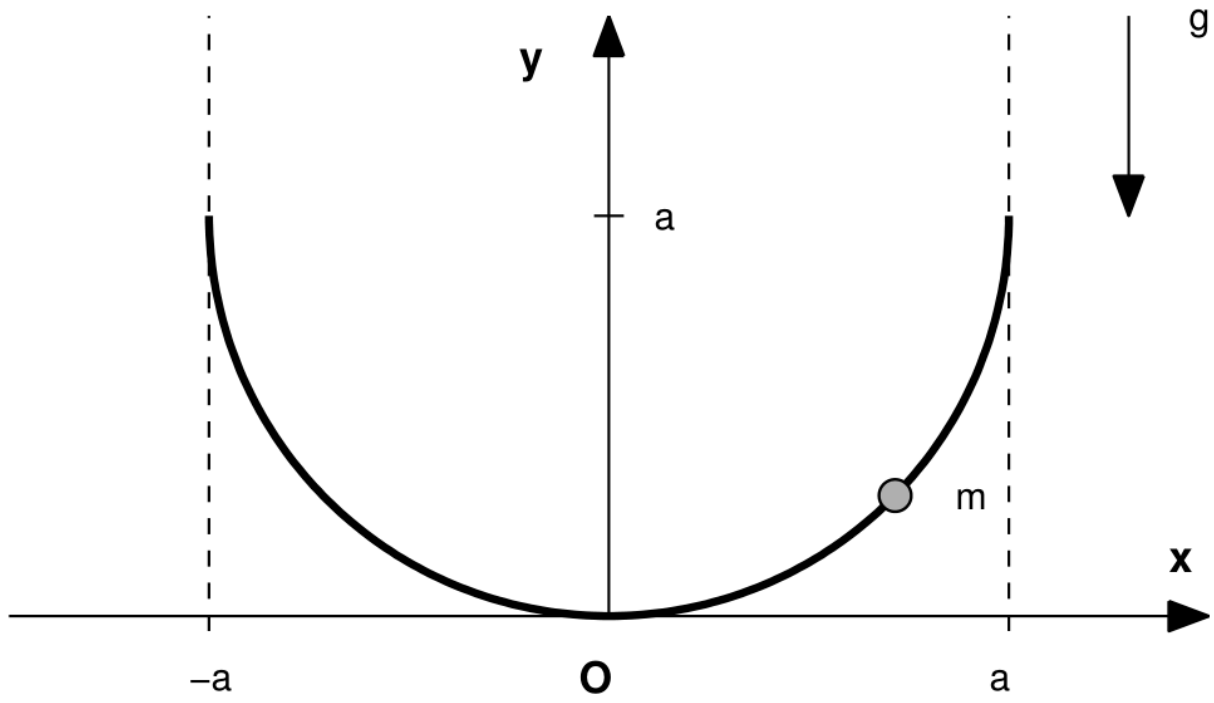


Figure: Half-circumference wire of radius a in the Oxy plane carrying the particle m, gravity along –y (Problems.pdf, p. 14).

(a) Determine the number of degrees of freedom of the problem.

(b) Define a convenient moving reference frame with origin in O, $OX_1Y_1Z_1$. Express the inertia force that acts on the particle in this moving reference frame.

(c) Express also the components of the weight of the particle with respect to $OX_1Y_1Z_1$.

(d) Determine if there are any equilibrium points.

(e) Reduce the problem to integrals (i.e. quadratures) and determine the value of the reaction force.

(f) Is it possible to use the energy equation? If Ω were a function of time, would it be possible to use the energy equation? Justify your answer.

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** A particle of mass m slides smoothly along a half-circumference wire of radius a lying in the X_1Y_1 plane, which rotates around the vertical OY axis with constant angular velocity $\boldsymbol{\omega} = \Omega\mathbf{j}$.
- **CDOFs:** 1 degree of freedom ($\theta \in [0, \pi]$).
- **Initial State:** At $t = 0, x_1(0) = 0, \theta(0) = 0$, velocity $\mathbf{v}_0 = v_0\mathbf{i}_1$.
- **Parameters:** $m, a, \Omega, g, v_0 > 0$.

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- Moving frame $\mathcal{F}_1 = \{O; \mathbf{i}_1, \mathbf{j}, \mathbf{k}_1\}$ rotating with $\boldsymbol{\omega}_{1/0} = \Omega\mathbf{j}$.
- Position on half-circle: $\mathbf{r}' = a\sin\theta\,\mathbf{i}_1 - a\cos\theta\,\mathbf{j}$.
- Polar tangent on wire: $\mathbf{e}_\theta = \cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{j}$.
- Relative velocity: $\mathbf{v}' = a\dot{\theta}\mathbf{e}_\theta$.
- **Change of basis $\mathcal{B}_0 \to \mathcal{B}_1$ (rotation about j).** $\mathcal{B}_1 = \{\mathbf{i}_1, \mathbf{j}, \mathbf{k}_1\}$ is the inertial basis $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ rotated by Ωt about $\mathbf{j}$, so $\mathbf{i}_1 = \cos\Omega t\,\mathbf{i} - \sin\Omega t\,\mathbf{k}$ and $\mathbf{k}_1 = \sin\Omega t\,\mathbf{i} + \cos\Omega t\,\mathbf{k}$:

$$[{}_0R_1] = \begin{pmatrix} \cos\Omega t & 0 & \sin\Omega t \\ 0 & 1 & 0 \\ -\sin\Omega t & 0 & \cos\Omega t \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\Omega t + \sin^2\Omega t = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

- **Change of basis $\mathcal{B}_1 \to \mathcal{B}_w$ (wire basis).** With $\mathbf{e}_r = \sin\theta\,\mathbf{i}_1 - \cos\theta\,\mathbf{j}$, $\mathbf{e}_\theta = \cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{j}$ and $\mathbf{e}_r \times \mathbf{e}_\theta = \mathbf{k}_1$, the triple $\{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}_1\}$ is right-handed:

$$[{}_1R_w] = \begin{pmatrix} \sin\theta & \cos\theta & 0 \\ -\cos\theta & \sin\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_1R_w] = \sin^2\theta + \cos^2\theta = 1, \qquad [{}_1R_w][{}_1R_w]^T = I$$

- Both matrices were checked symbolically with sympy ($\mathbf{i}_1 \times \mathbf{j} = \mathbf{k}_1$ included).

PHASE 3

Step-by-Step Mathematical Deduction

Part (a): Degrees of Freedom

- Ambient space: 3 coordinates. Two constraints: wire geometry ($r = a$, $z_1 = 0$).
- Total CDOFs: $3 - 2 = 1$.

Part (b): Inertia Forces in $\mathcal{F}_1$

1. Centrifugal Force:

$$\mathbf{F}_{cf} = m\Omega^2 x_1 \mathbf{i}_1 = m\Omega^2 a \sin\theta\,\mathbf{i}_1$$

2. Coriolis Force:

$$\mathbf{F}_{Cor} = -2m(\Omega\mathbf{j}) \times (a\dot{\theta}\mathbf{e}_\theta) = 2m\Omega a\dot{\theta}\cos\theta\,\mathbf{k}_1$$

Part (c): Components of Weight

$$\mathbf{F}_g = -mg\mathbf{j}$$

Part (d): Equilibrium Points in $\mathcal{F}_1$

Project equation of motion along the wire tangent $\mathbf{e}_\theta$:

$$ma\ddot{\theta} = (\mathbf{F}_g + \mathbf{F}_{cf}) \cdot \mathbf{e}_\theta = (-mg\mathbf{j} + m\Omega^2 a \sin\theta \mathbf{i}_1) \cdot (\cos\theta\, \mathbf{i}_1 + \sin\theta\, \mathbf{j})$$

$$ma\ddot{\theta} = m\Omega^2 a \sin\theta\cos\theta - mg\sin\theta = ma\left(\Omega^2\cos\theta - \frac{g}{a}\right)\sin\theta$$

Equilibria ($\ddot{\theta} = 0$):

1. $\sin\theta = 0 \implies \theta = 0$ (lowest point, stable if $\Omega^2 < g/a$).

2. $\cos\theta = \frac{g}{\Omega^2 a}$. This requires $\Omega^2 > g/a$, yielding the stable equilibrium $\theta_0 = \arccos\left(\frac{g}{\Omega^2 a}\right)$.

Part (e): Reduction to Quadratures & Reaction Force

Effective potential:

$$V_{eff}(\theta) = -mga\cos\theta - \frac{1}{2}m\Omega^2 a^2 \sin^2\theta$$

Energy conservation:

$$\frac{1}{2}ma^2\dot{\theta}^2 + V_{eff}(\theta) = E' = \frac{1}{2}mv_0^2 - mga$$

$$\dot{\theta} = \pm\sqrt{\frac{v_0^2}{a^2} + \frac{2g}{a}(\cos\theta - 1) + \Omega^2 \sin^2\theta}$$

Out-of-plane reaction:

$$N_z = -F_{Cor,z} = -2m\Omega a\dot{\theta}\cos\theta$$

Part (f): Energy Equation Validity

Energy is conserved in $\mathcal{F}_1$ because the rotation rate Ω is constant. If $\Omega = \Omega(t)$, Euler forces do work and $\partial V_{eff}/\partial t \neq 0$, breaking energy conservation.

PHASE 4

Physical Interpretation & Dimensional Verification

- Supercritical pitchfork bifurcation at $\Omega_c = \sqrt{g/a}$.
- **Dimensional check:** $[m\Omega^2 a] = \mathrm{kg}\cdot\mathrm{s}^{-2}\cdot\mathrm{m} = \mathrm{N}$ (centrifugal force); $[\Omega^2\cos\theta - g/a] = \mathrm{s}^{-2}$; $[N_z] = [m\,\Omega\,a\,\dot{\theta}] = \mathrm{N}$; in the expression of $\dot{\theta}$ every term has units s^{-2} under the root.
- **Limits:** for $\Omega \to 0$ the effective potential reduces to $-mga\cos\theta$ (simple pendulum with a single stable equilibrium at $\theta = 0$) and $N_z \to 0$; at $\Omega = \Omega_c = \sqrt{g/a}$ the off-axis equilibrium $\cos\theta_0 = g/(\Omega^2 a) \to 1$ merges with $\theta = 0$; for $\Omega \to \infty$, $\theta_0 \to \pi/2$.
- **Numerical check ($\Omega^2 = 2g/a$):** $\cos\theta_0 = 1/2$, $\theta_0 = 60^\circ$; $V''_{eff}(\theta_0) = mga\cos\theta_0 - m\Omega^2 a^2\cos 2\theta_0 = 1.5\,mga > 0$ (stable) and the small-oscillation frequency is $\omega^2 = V''_{eff}/(ma^2) = 1.5\,g/a = \Omega^2\sin^2\theta_0$, which agree.

Source: Problems.pdf (Page 14)